

called an ultrapower of the set A . In general the construction is used with a free ultrafilter.

ULTRASONIC IMAGING (SONOGRAPHY; ULTRASONOGRAPHY): The use of ultrasound (waves above 20 KHz and not detectable by human ear) to visualise a recording of the reflections (echoes) of objects such as a metallic or biological object that is opaque to light by recording the reflections (echoes) waves directed into the tissues. Ultrasound pulses are transmitted into the body and are reflected at boundaries between different tissue types back to the transducer as echoes. The echoes are then converted into electrical impulses that are displayed as still images or as a moving picture on an oscilloscope, displaying an image of the tissues under examination. Ultrasound can be used to monitor the function of organs or blood flow as in Doppler Effect, where the transmitter is separate from the receiver. It is also delivered at large intensities to localised regions of the body for therapeutic purposes.

ULTRASONICS: The study and use of pressure waves in elastic media that have a frequency above 20 kilohertz and are therefore inaudible to the human ear. Ultrasonic waves can be generated or detected by piezoelectric transducers. It is used in medicine for diagnosis, especially in conditions like pregnancy, where x-ray would have harmful effects. It is also used to break kidney stones, to test flaws in metals, ultrasonic welding, drilling, etc.

ULTRASOUND (ULTRASONIC SOUND): Waves of higher frequency above 20 KHz that the human ear cannot detect and used in ultrasonic imaging.

ULTRASOUND SCAN: A medical imaging technique in which high frequency (MHz) sound waves are transmitted and reflected from soft tissue. It is used to view soft tissues such as internal organs and body cavities. It can also be used to examine a foetus during pregnancy.

ULTRASTRUCTURE (FINE STRUCTURE): The sub-microscopic almost molecular, structure of a biological specimen such as a cell, tissue or organ which is revealed by the use of an electron microscope.

ULTRAVIOLET (UV): Of or relating to electromagnetic radiation that has a wavelength shorter than that of the violet end of the visible spectrum but longer than that of X-rays. It has frequencies between 790 terahertz and 30 petahertz and wavelengths between 380 and 10 nanometers. UV light is invisible to the human eye. The main source of UV is the Sun; some artificial light such as from tanning and electric arc lamps also radiate UV. UV radiation can burn the skin and cause skin cancer. However, it has some health benefits, because when it hits human skin, it triggers the production of vitamin D, which promotes the growth of bones and teeth.

ULTRAVIOLET ASTRONOMY: The branch of astronomy that studies cosmic ultraviolet emissions from sources such as stars, planets, galaxies and interstellar gas clouds.

ULTRAVIOLET MICROSCOPE: A microscope that uses ultraviolet radiation to illuminate the specimen and form the image. This increases the resolution because ultraviolet radiation has much shorter wavelengths (about 300 nm) than visible light and resolution increases as wavelength decreases. Ultraviolet microscopy is difficult and complex and is rarely used since the development of the electron microscope.

ULTRAVIOLET RADIATION (UV RADIATION): Electromagnetic radiation having wavelengths between 4–400 nanometres, beginning at the short-wavelength limit of visible light and overlaps the wavelengths of long x-rays. In the range 400–300 nm the radiation is known as the **near ultraviolet**. In the range 300–200 nm it is known as the **far ultraviolet**. Below 200 nm it is known as the **extreme ultraviolet** or the **vacuum ultraviolet**, as absorption by the oxygen in the air makes the use of evacuated apparatus essential. The Sun is a strong emitter of UV radiation, but only the near UV reaches the surface of the Earth as the ozone layer of the atmosphere absorbs all wavelengths below 290 nm. Most UV radiation for practical use is produced by various types of mercury-vapour lamps. The longest-wavelength UV radiation is not harmful in normal doses and is used clinically in the treatment of certain skin complaints, such as psoriasis and also to induce vitamin D formation in patients who are allergic to vitamin D preparations. It is also used to treat jaundice in newborns

and to sterilise equipment. UV-B (290–320 nm) causes reddening of the skin followed by pigmentation (tanning). UV-C (280–100 nm) has the shortest wavelengths and this causes skin cancer. The risk of contracting this has increased by the depletion of the ozone layer.

ULTRAVIOLET-VISIBLE SPECTROSCOPY (ULTRAVIOLET-VISIBLE SPECTROPHOTOMETRY; UV-VIS, UV/VIS): A technique used for the quantitative determination of solutions of transition metal ions and highly conjugated organic compounds. The absorption in the visible range directly affects the perceived colour of the chemicals involved. The electronic transitions in molecules occur in the visible and ultraviolet regions of the electromagnetic spectrum and that a given transition occurs at a characteristic wavelength. The spectrometer has two sources, ultraviolet and white visible light which together cover the whole wavelength of the instrument.

ULURU (AYERS ROCK): The largest individual rock mass in the world with a height of 868 metres. It is located in the south of the Northern Territory, Australia. Uluru is listed as a World Heritage Site.

ULVALES: An order of the Chlorophyceae containing pseudoparenchymatous thalloid algae (e.g. *Ulva*), characteristically found in marine and brackish waters. The sheet-like thallus is formed by division in two planes of a filament that initially resembles those of the Ulotrichales. The Ulvales are consequently often placed in this order. There is an isomorphic alternation of generations in some species.

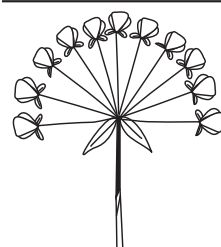
UMBEL: An umbrella-shaped inflorescence in which the pedicels of the flowers are equal in length and borne on short stems arising from the top of a main stem. This inflorescence type is typical of members of the Apiaceae. Examples are parsley, carrot, dill, anise, caraway, celery and fennel.

UMBELLIFERAE (APIACEAE): A dicotyledonous family containing about 2500–3000 species in about 300 genera and commonly known as the carrot family. Its members are distinguished by their characteristic inflorescence, the umbel, a feature that made them one of the first natural families to be clearly recognised.

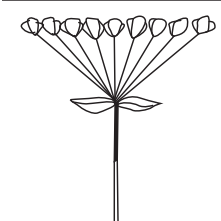
UMBELLIFORM: Resembling or in the form of an umbel, for example, an inflorescence with the general appearance, but not necessarily the structure, of a true umbel. The term is often applied to inflorescences which are condensed cymes rather than condensed racemes.

UMBILICAL CORD: 1. The flexible, often spirally twisted tube, about 56 centimetres in length in the pregnant human female that connects the embryo to the placenta. It carries nourishment and oxygen from the placenta to the foetus and return waste products to the placenta from the foetus. It consists of a continuation of the membrane covering the foetus and encloses a mucoid jelly through which one vein carries oxygenated blood and two arteries carry de-oxygenated blood. It is severed after birth to free the newly born baby from the placenta and shrivels to leave a scar called the navel. 2. A cable, tube or pipe such as the tube that attaches somebody or something to an essential supply, for example, the tube that connects a deep-sea diver to oxygen supply on board a ship

UMBILICAL POINT (UMBILIC): In Euclidean geometry, a point on a surface that is either planar (when normal curvature vanishes) or circular (when the principal radii of normal curvature are equal). The sphere is the only surface with non-zero curvature where every point is umbilic. A flat umbilic is an umbilic with zero Gaussian curvature. The three main type of umbilic points are elliptical umbilics, parabolic



Umbel, round



Umbel, flat



Umbel, compound

umbilics and hyperbolic umbilics. Elliptical umbilics have the three ridge lines passing through the umbilic and hyperbolic umbilics have just one. Parabolic umbilics are a transitional case with two ridges one of which is singular. All points where an ellipsoid of revolution cuts its axis of revolution are umbilical.

UMBILICUS: A nave-like structure or a small opening or depression similar to a navel, such as the hilum of a seed, the hollow at the base of the shell of some gastropod molluscs or one of the openings in the shaft of a feather.

UMBO: A blunt or rounded protuberance arising from a surface, such as on the ends of the scales of some pine cones or the prominence near the hinge of a bivalve shell. **Umbonate** means possessing umbos.

UMBRA (FULL SHADOW): 1. The region of a shadow that is totally dark because no light reaches it and from which no part of the light source can be seen. The shadow from a point source of illumination is essentially all umbra, but that from a source of some size such as the Sun consists of a small umbra and a much larger partial shadow called the **penumbra**. In an eclipse of the Sun, the regions within the umbra experience a total eclipse and those within the penumbra have partial eclipse. 2. The inner, darker area of a sunspot.

UMBRACULIFEROUS: Bearing an open umbrella-shaped structures. **Umbaculate** means umbrella-shaped.

UMPIRE SAMPLE: A sample taken, prepared and stored in an agreed upon manner for the purpose of settling a dispute. The agreement usually extends beyond the sample to the basis for reaching a decision, for example, quantity of material from which taken, use of a third party and criteria serving as the basis for acceptance, rejection or economic adjustment.

UMTS (UNIVERSAL MOBILE TELECOMMUNICATIONS SYSTEM): A third generation (3G) of mobile phone networks based on the GSM standard that transmits broadband data at rates of up to 2 megabits per second (Mbps) or more.

UNARMED: In botany, it refers to structures or parts without spines, prickles or thorns.

UNARY (MONADIC; SINGULAR): Of an operator, predicate, etc., having only a single argument place. Unary operation is a rule that associates with any element of S a resulting element so that the resulting element is always also in S, then it is said that S is closed under the operation. Examples are the rule that associates with each integer a its negative $-a$; the rule with each non-zero real number a its inverse $-$; and the rule that associates with any subset A of a universal set ϵ its complement A'. Negation, inversion and set complementation are monadic operators. A monadic relation is a (one place) predicate.

UNBALANCED FORCES: Forces that are not equal and opposite in direction. When unbalanced forces act on an object at rest, the object will move. Also, when unbalanced forces act on a moving object, the velocity of the object will change (it changes the position, speed or direction of the object to which it is applied). The direction of the object is determined by the bigger force. Examples of unbalanced forces include the force of gravity and applied forces.

UNBIASED: 1. In statistics, an unbiased sample is one that is not affected by any extraneous factors, conflated variables or selectivity that influence its distribution and can be an accurate representation of the entire population that can help draw conclusions about the population. An unbiased estimate of a given parameter is having the mean of the sampling distribution of that statistic being equal to the parameter being estimated; having zero bias. 2. Unbiased significance test is having a power greater than the predetermined significance level.

UNBOUNDED: 1. An unbounded set is one that is not bounded; a set that lacks either a lower bound or an upper bound. For example, the sequence 1, 2, 3, 4, ... is unbounded, because its upper and/or lower bounds are not finite. 2. Of a function on a set, having values that increase without bound in modulus or norm for arguments in

the given set; that is, for all large N there is a value of the function of which the modulus or norm is greater than N. If measure is involved, an unbounded function is one that is not essentially bounded.

UNCERTAINTY (INFORMATION): A real-valued function of events in a probability space that depends only on the probability of events. It is such that events of probability one have zero that increases as probability drops and that uncertainty simultaneous occurrence of two independent events is the sum of their individual uncertainties.

UNCERTAINTY OF MEASUREMENT: Parameter, associated with the result of a measurement that characterises the dispersion of the values that could reasonably be attributed to the measurand (the quantity being measured).

UNCERTAINTY PRINCIPLE (HEISENBERG'S UNCERTAINTY PRINCIPLE; PRINCIPLE OF INDETERMINISM): A principle in quantum mechanics holding that it is impossible to determine with unlimited accuracy both the position and momentum of a particle at the same time. It is stated in the form; $\Delta x \Delta p_x \geq h/4\pi$, where Δ_x is the uncertainty in the x-coordinate of the particle, Δp_x is the uncertainty in the x-component of the particle's momentum and h is the Planck constant. An explanation of the uncertainty principle is that in order to locate a particle exactly, an observer must be able to bounce a photon of radiation off the particle. This act of location itself alters the position of the particle in an unpredictable way. To locate the position of a particle accurately, photons of short wavelength would have to be used. The high momenta of such photons would cause a large effect on position. Using photons of lower momenta would have less effect on the particle's position, but would be less accurate because of the longer wavelength.

UNCINATE: Hooked at the tip.

UNCOMMITTED LOGIC ARRAY, ULA (GATE ARRAY): In computer science, it is a type of partially finished integrated circuit in which the logic gates are laid down to a general-purpose design but are not connected to each other. The interconnections can then be set in place according to the requirements of individual manufacturers. Producing ULAs may be cheaper than using a large number of TTL (transistor-transistor logic) chips or commissioning a fully customised chip.

UNCONDITIONAL CONVERGENCE: In mathematics, the requirement that a series converge in any reordering of the terms; for a complex series this coincides with absolute convergence. For example, $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ is

unconditionally convergent, as it is absolutely convergent while $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ is not.

UNCONDITIONAL INEQUALITY: In mathematics, an inequality that is universally true and holds true for any values assigned to the variables or which contains no variables. For example, $x + 1 > x$ is an unconditional inequality, while $x^2 > x$ is a conditional inequality.

UNCORRELATED VARIABLES: Variables that display zero correlation.

UNCOUNTABLE SET (UNCOUNTABLY INFINITE SET): A set of numbers that cannot be put in a definite order from the smallest to the largest and so cannot be counted. In other words, an uncountable set contains so many elements that they cannot be put in one-to-one correspondence with the set of natural numbers. For example, the set of real numbers in the interval [0,1] is uncountable. All uncountable sets are infinite, but not all infinite sets are uncountable. The best known uncountable set is the set of all real numbers. By contrast the set of all natural numbers, which represents the smallest type of infinity, is countable, for example, {0, 1, 2, 3, ...}.

UNCOUPLER: A substance that inhibits adenosine triphosphate (ATP) synthesis and enhances phosphorylation of adenosine diphosphate (ADP) from electron transfer. An example is 2, 4- dinitrophenol.

UNCOUPLING PROTEIN (UCP): Any of a family of proteins that uncouple electron transport from oxidative phosphorylation in mitochondria so that the energy derived from aerobic respiration is released chiefly as heat instead of being used to make adenosine triphosphate (ATP). There are five types of UCP: UCP1, also known as thermogenin; UCP2; UCP3; SLC25A27, also known as "UCP4"; and SLC25A14, also known as "UCP5". The archetype is the UCP found in brown fat, a type of adipose tissue associated with newborn or hibernating mammals.

UNDECAGON: A two-dimensional geometric figure formed of eleven angles and eleven sides or an 11-sided polygon.

UNDECIDABLE: 1. In logic, of a formal system, having the property that it does not have a decision procedure; not decidable. 2. Of a formula or theorem, neither it nor its negation is provable within the axioms of the given theory and therefore not considered as a theorem. 3. In computer science, an **undecidable problem** is a decision problem that admits no algorithmic solution. A decision problem is a problem that always requires a yes-or-no answer and since it is impossible to construct an algorithm for its solution, it is undecidable. An undecidable problem can never be solved by any computer system that always leads to a correct yes-or-no answer.

UNDEFINED ELEMENT: A non-logical constant, a primitive member of some axiomatically defined structure, for example, points and lines are undefined elements in axiomatic geometry and ε is undefined predicate in set theory.

UNDEFINED SLOPE: The slope of a vertical line, where the amount of vertical change equals a non-zero real number, and the amount of horizontal change equals zero, yielding a slope ratio whose denominator equals zero.

UNDERLYING FUNCTOR (FORGETFUL FUNCTOR; STRIPPING FUNCTOR): A functor obtained by considering a category of algebraic forms such as groups, Abelian groups, modules, rings and vector spaces, as another category of sets with simpler objects. For example, the functor from the category of groups to the category of sets that preserves all mappings forgets the structure of the groups.

UNDERLYING SET: In mathematics, it is the set on which a topology or other structure is defined.

UNDERNOURISHMENT: The outcome of insufficient food intake and repeated infectious diseases. It includes being underweight for one's age, too short for one's age (stunted), dangerously thin for one's height (wasted) and deficient in vitamins and minerals (micronutrient malnutrition).

UNDERDETERMINED: In mathematics, of a system of linear equations or a system of polynomial equations, involving fewer equations than variables. An example of underdetermined equation is a system with two equations and three unknowns. An underdetermined linear system is either consistent or inconsistent, depending on the equations and has either no solution or infinitely many solutions. For example, the following system of equations is not consistent and has no solution: $x + y + z = 1$ and $x + y + z = 0$.

UNDETERMINED: In mathematics, of a parameter of an unspecified form; for example, constants of integration are undetermined.

UNDETERMINED ELEMENT: A non-logical constant, a primitive member of some axiomatically defined structure. For example, points and lines are undefined elements in axiomatic geometry and ' ε ' is an undefined predicate in set theory.

UNDIFFERENTIATED SOIL MAP UNIT: Soil mapping unit in which two or more soil units occur, but not in a regular geographic association. For example, the steep phases of two or more soils might be shown as a unit on a map because slope is the prime characteristic.

UNDULATING NUMBER: An integer whose digits, in a given base, alternate, one is written in the form $ababab\dots$, where a and b are digits. For example, 434343 and 101010101 are undulating numbers.

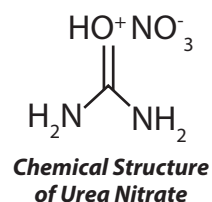
UNDULIPODIUM: A term that refers to both cilia and flagellum in

eukaryotes. They are used mainly for locomotion; other functions include creating feeding currents and sweeping debris off the epithelial surfaces. They are supported by microtubules.

UNDULOID: A member of a family of curves that is formed by films or liquid drops suspended between certain boundaries. Examples of unduloids are seen on a spider web when viewed through a microscope. They consist of blobs of viscous liquid, which make up the sticky part of the web and are mostly gathered into a lemon shape.

UNEQUAL CLEAVAGE: A holoblastic cleavage producing blastomeres of unequal size. An example is the holoblastic cleavage in a frog egg, where a fair amount of yolk concentrates toward the vegetal pole. Because of the concentration of the yolk at the vegetal pole, cleavage results in blastomeres in different size. That is, the blastomeres are smaller in the animal pole while larger in vegetal pole.

UNFO (UREA NITRATE-FUEL OIL): Colourless, explosive, fire-hazardous crystal with the chemical formula $\text{CH}_5\text{N}_3\text{O}_4$, which decomposes at 152 °C (425 K). It is soluble in alcohol, slightly soluble in water. It is used in explosives and to make urethane. It is produced by nitration of urea using nitric acid similar in action to ammonium nitrate-fuel oil (ANFO). It has a destructive power similar to better-known ammonium nitrate-fuel oil (ANFO) explosives, with a velocity of detonation between 3,400 m/s and 4,700 m/s.



UNGUICULATE: Clawed or having talon-like extensions.

UNGULATE: A hoofed, herbivorous, quadruped, placental mammal such as camel, deer and horse. Ungulates are grouped into three or four orders: (i) Artiodactyla, the even-toed ungulates including pigs, camels, deer, bovines, etc; (ii) Perissodactyla, the odd-toed ungulates including horses, tapirs, rhinoceroses, etc; and (iii) Proboscidea, the elephants.

UNGULIGRADE: The disposition of standing on hooves of ungulates, in which only the tips of the digits are on the ground and the rest of the feet are off the ground.

UNI: A prefix meaning one.

UNIAXIAL: 1. Referring to a plant with a single unbranched stem that terminates in a flower. 2. Referring to crystals having one direction along which double refraction of light does not take place.

UNIAXIAL CRYSTAL: A double-refracting crystal having only one optic axis along which light can propagate without exhibiting double refraction.

UNIAXIAL PRESSING: Application of pressure in one direction during ceramisation to achieve a uniform densification and the production of a compact monolith.

UNIAXIAL SAMPLE: Sample characterised by a (unique) sample axis Z with all directions perpendicular to Z being equivalent. In other words, the sample properties are invariant to rotation around Z . Uniaxiality exists in many anisotropic samples and simplifies the interpretation of their spectra considerably.

UNICARPELLOUS: Having a single, free carpel.

UNICAST: A type of communication in which data is sent from just a single source to a specified destination. It a form of transmission used mostly on LANs such as Ethernet and within the Internet.

UNICELLULAR: Referring to organisms made up of or consisting of only a single living cell, such as the protozoans. Examples are bacteria, protozoa and certain algae.

UNICITY (UNIQUENESS THEOREM): The unicity theorem is the uniqueness of a mathematical object, which usually means that there is only one object fulfilling given properties or that all objects of a given class are equivalent (i.e., they can be represented by the same model). The object of many uniqueness theorems is the solution to a problem or an equation; in such cases, a uniqueness theorem is normally combined with an existence theorem.

UNICODE: A set of universal codes used to represent letters, numbers, control characters and the like, designed for use in computers. It defines the way individual characters are represented in text files, web pages and other types of documents. It is capable of having a unique number between U+0000 to U+10FFFF, which means a Unicode may be 8-bit, 16-bit or 32-bit. Each character is assigned a code point, for example, U+0041 is the English letter A. While American Standard Code for Information Interchange (ASCII) represents only basic English characters, Unicode supports characters from all languages around the world. ASCII character set only supports 128 characters, while Unicode can support roughly 1,000,000 characters. ASCII uses one byte to represent each character and Unicode supports up to 4 bytes for each character.

UNICODE TRANSFORMATION FORMAT (UTF): A term used to refer to several types of Unicode character encodings, including UTF-7, UTF-8, UTF-16 and UTF-32.

UNICOSTATE: Having a single main costa, rib or riblike part, for example, a unicastate leaf.

UNICURSAL: Relating to or denoting a curve or surface that is closed but not having any part retraced; it can be drawn or swept out in a single movement.

UNIDENTIFIED FLYING OBJECT (UFO): Any unusual apparent object or phenomenon in the sky whose cause cannot be identified by the observer, especially those considered to have extraterrestrial origins.

UNIFIED ATOMIC MASS UNIT: Non-SI unit of mass, which is equal to the atomic mass constant and defined as one twelfth (1/12) of the mass of a carbon-12 atom in its ground state. It is used to express masses of atomic particles, $u \approx 1.6605402(10) \times 10^{-27}$ kg.

UNIFIED MODEL: 1. Also known as **collective model**, it is a single model of the atomic nucleus that combines the shell model with the liquid-drop model (the single-particle and collective aspect of nuclei) to explain certain magnetic and electric properties that neither of the two separately can explain. In the shell model, nuclear energy levels are calculated on the basis of a single nucleon (proton or neutron) moving in a potential field produced by all the other nucleons. Nuclear structure and behaviour are then explained by considering single nucleons beyond a passive nuclear core composed of paired protons and paired neutrons that fill groups of energy levels. 2. A large and complex numerical model used in numerical weather prediction and studies of climate change originally developed by the United Kingdom Meteorological Office.

UNIFIED MODELLING LANGUAGES: A general purpose graphical language used in the representation of systems at the design stage. It employs three categories of diagrams: structural, behavioural and model management.

UNIFIED-FIELD THEORY: A comprehensive theory that relates the basic identity of electricity and magnetism and the standard model postulates a basic identity for the strong force, the weak force and electromagnetism. Unified field theories and other fundamental theories, such as superstring theory and super membrane theory, are of great importance in understanding cosmology, particularly the early universe.

UNIFLOROUS: Bearing only a single flower.

UNIFOLIOLATE: A compound leaf which has been reduced to a single, usually terminal leaflet; or a leaf, which is compound in structure, but with only a single leaflet that appears simple, for example, the leaves of the citrus plant. **Unifoliate** means with a single leaf.

UNIFORM: 1. Same, not varying in form, quality, texture, size, etc. 2. Conforming to one principle, standard or rule; consistent. 3. Being the same as or consonant with another or others.

UNIFORM ACCELERATION: Acceleration is said to be uniform if a body covers equal velocity in equal time intervals. It is calculated as Uniform acceleration = final velocity - initial velocity/time taken

$$a = \frac{v_f - v_i}{t}$$

UNIFORM BOUND: In mathematics, a bound that holds uniformly, usually for a set of functions. Bounded functions are functions for which there exists a lower bound and an upper bound, in other words, a constant that is larger than the absolute value of any value of this function.

UNIFORM BOUNDED PRINCIPLE (BANACH-STEINHAUS THEOREM): The theorem that a pointwise-bounded family of continuous linear operators between a Banach space and a normed space is equicontinuous (uniformly bounded), if for each x in the unit ball, $\sup_i \|T_i(x)\|$ is finite, then actually $\sup_i \|T_i\|$ is finite.

UNIFORM BOUNDEDNESS PRINCIPLE (BANACH-STEINHAUS THEOREM): The theorem that a pointwise-bounded family of continuous linear operators between a Banach space and a normed space is equicontinuous (uniformly bounded): if for each x in the unit ball, $\sup_i \|T_i(x)\|$ is finite, then actually $\sup_i \|T_i\|$ is finite.

UNIFORM CIRCULAR MOTION: A motion of a body along a circular path. The speed of the body in circular motion remains constant, but its direction changes continuously so the velocity changes and hence the body accelerates. Since this acceleration is strictly a change in direction and is a result of a force directed toward the centre of the circle, the force is a centripetal force, causing centripetal acceleration. An object moving in a circular motion around another object is said to be orbiting the object. For example, the Earth orbits around the Sun from an average distance of 149.59787 million kilometres in 365 days.

UNIFORM CONTINUITY: In mathematics, the property of a function between metric spaces that continuity is uniform, so that for every $\epsilon > 0$ there is a single $\delta > 0$ such that $d(f(x), f(y)) < \epsilon$ whenever $d(x, y) < \delta$, whereas in general δ would depend on both ϵ and x . If the domain is compact uniform continuity follows automatically from continuity.

UNIFORM CONVERGENCE: In mathematics, the property that all of a family of functions or series on a given set converge at the same rate throughout the set; that is, for every $\epsilon > 0$ there is a single N such that for all points in the set, $|f_m(x) - f_n(x)| < \epsilon$ for all $m, n > N$ and similarly for uniform convergence as x tends to a value a . This is convergence in the uniform or Chebyshev norm. A power series converges uniformly within any disk of strictly smaller radius than its radius of convergence. The uniform limit of a sequence of continuous functions is continuous, as opposed to the limit in pointwise convergence that may well not be. This is often the case for Fourier series.

UNIFORM CONVEXITY (UNIFORM ROTUNDITY): The property of a norm or ball that for every $\epsilon > 0$ there is a single $\delta > 0$ such that $\|(x - y)\| < \epsilon$ whenever $\frac{x+y}{2} > 1 - \delta$, and x and y have norm

one. A normed space on which such an equivalent norm may be placed is super-reflexive, a stronger property than reflexivity, possessed by L_p spaces for $1 < p < \infty$. In finite dimensions such norms coincide with strictly convex ones. Local uniform convexity arises if x is fixed throughout the definition.

UNIFORM CORROSION: Corrosion is uniform if the time average of the corrosion current through a unit area of any macroscopic dimension is independent of the position on the surface.

UNIFORM DISTRIBUTION: 1. The property of an infinite sequence of numbers $\{a_n\}$ in an interval $[a, b]$ that, for any $a < \alpha < \beta < b$, the probability that the numbers lie in $[\alpha, \beta]$ is the ratio precisely, one requires that this ratio is equal to $\frac{\beta - \alpha}{b - a}$; precisely, one requires

that this ratio is equal to $\lim_{N \rightarrow \infty} \frac{\{n < N : a_n \in [\alpha, \beta]\}}{N}$. 2. In statistics,

the distribution of a random variable on an interval $[a, b]$ when its probability density function is zero outside this interval and is $1/(b - a)$ at any point of the interval. Its mean is the midpoint of the interval and its variance is $(b - a)^2/12$. The distribution associated with normalised Lebesgue measure on a set in n -space, all sets of the same size being equally likely.

UNIFORM FIELD: A field in which the value of the field strength is the same at all points at the instant under consideration. For example, a uniform electric field exists between two parallel charged plates. Uniform magnetic fields may be produced by Helmholtz coils.

UNIFORM MINIMUM VARIANCE: The property of a statistic of having the minimal variance of all statistics of a certain class, often the class of unbiased estimators, for every value of the parameters.

UNIFORM POLYHEDRON: A polyhedron in which each face is regular and each vertex is equivalently arranged. Uniform polyhedra include the Platonic solids, the Archimedean solids, the prisms and antiprisms and the nonconvex uniform polyhedra.

UNIFORM RATE: Constant rate, for example, the price of a piece of cloth per metre, so that the length of the cloth bought is multiplied by the rate. Non-uniform rate is where the rate is not constant, for example, income tax.

UNIFORM RESOURCE IDENTIFIER (URI): A string of characters that identifies the name and location of a file or resource in a uniform format. URIs provide a standard way for resources to be accessed by other computers across a network or over the World Wide Web. They are used by software programs such as Web browsers and P2P file-sharing programs to locate and download files. The most common form of URI is the uniform resource locator (URL)

UNIFORM RESOURCE LOCATOR, URL (WEB ADDRESS): A standardised naming convention for addressing a specific Web site or file on the Internet. It usually consists of the access protocol (http), the domain name (www.hmco.com) and optionally the path to a file or resource residing on that server (trade). It does not have spaces or certain other characters and uses forward slashes to denote different directories. Examples of URLs are <http://www.bro.com/>, <http://web.ust.edu/> and <ftp://info.apple.com/>. Not all URLs begin with "http" and some portions of the URL are not required and many times can be left out.

UNIFORM SPACE: A topological space S having an additional structure that is used to define uniform properties such as completeness, uniform continuity and uniform convergence. The topology is induced by a uniformity (a class of subsets of $S \times S$) such that U is open if and only if for any $x \in U$ there exists V in the uniformity with $\{y : (x, y) \in V\}$ contained in U .

UNIFORM SUBSTITUTION: The replacement of every occurrence of some well-formed part of a given expression by some other expression, to yield a substitution instance.

UNIFORMITARIANISM (PRINCIPLE OF UNIFORMITARIANISM): The idea that physical and chemical geological processes have remained uniform over time and that slight changes over long periods can have large-scale consequences. This was proposed by James Hutton in 1795 and refined by Charles Lyell during the 1800s. The principle, expressed as 'the present is the key to the past', is fundamental to geologic thinking and explains that processes operating today on Earth operated in much the same way in the geologic past.

UNIFORMITY: In mathematics, uniformity on a set S is a filter of subsets F of $S \times S$ such that each member of the class contains all points of the form (x, x) , such that if V lies in F so does the inverse $v^{-1} = \{(x, y) : (y, x) \in V\}$ and such that for each V in F there exists a W in F with the composition

$W \circ W = \{(x, z) : (x, y) \in W, (y, z) \in W\}$ lying in V . Each element of the uniformity is called a vicinity. Every metric space is a uniform space.

UNIFORMLY BOUNDED: A set of functions having a uniform bound, that is if there exist one single constant number which bounds all the functions of the given set.

UNIFORMLY CONTINUOUS: For a function f and a subset M of its domain, f is uniformly continuous on M if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all a, x from M satisfying $|x - a| < \delta$, there is $|f(x) - f(a)| < \varepsilon$.

UNIFORMLY LABELLED: A selectively labelled compound, where all atoms of a particular element are labelled in the same isotopic ratio.

UNIFORMLY LABELLED TRACER: A tracer in which the label is uniformly distributed over its possible positions.

UNIUGATE: A leaf, which is pinnately compound, but consisting of one pair of leaflets.

UNILATERAL: 1. In botany, the term refers to one-sided, as in inflorescence with the flowers all on one side of the axis. 2. A **unilateral surface** is a surface with only one side, such as a Moebius strip and the Klein bottle. 3. Of or denoting a limit that is evaluated on one side of the limiting value of the independent variable. **Unilateral analysis** is the study of one-sided properties such as lower semi-continuity, convexity and minimality.

UNILATERAL SURFACE: A surface with only one side, such as a Moebius strip and the Klein bottle.

UNILOCULAR: With a single locule or compartment, as occurs in some ovaries.

UNILOCULAR SPORANGIUM: A sporangium that is not divided and comprises only one locule or compartment. Unilocular sporangia are seen in the sori of the brown alga *Laminaria*.

UNIMODAL: 1. Of a real function f , defined on an interval possessing a single maximum or minimum in the interval; that is, there is a unique point c in the given interval, $[a, b]$, such that the function is monotone in opposite senses on $[a, c]$ and on $[c, b]$. Thus either if $x < y < c$ then $f(x) < f(y)$, if $x > y > c$ then $f(y) > f(x)$ or vice versa. 2. In statistics, it is a unimodal probability distribution or a probability distribution with only one mode or with a single clear peak.

UNIMODULAR: A square integer matrix with determinant of ± 1 . Equivalently, it is an integer matrix that is invertible over the integers: there is an integer matrix N which is its inverse (these are equivalent under Cramer's rule). Thus every equation $Mx = b$, where M and b are both integer and M is unimodular, has an integer solution. The unimodular matrices of order n form a group, which is denoted by $GL_n(\mathbb{Z})$.

UNIMODULAR MATRIX: A square integer matrix having a determinant of ± 1 .

UNIMOLECULAR REACTION: A chemical reaction that involves only one type of molecular species as the reactant, for example, the decomposition of mercury(II) oxide, HgO , on strong heating to give mercury and oxygen. Another example is $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$, as in the electrolytic dissociation of water.

UNINSTALL: In computing, a term used to show steps to safely and efficiently removing an installed program and all associated files on the computer.

UNINTERPRETED: Of a formal theory, considered solely in terms of its syntactic structure rather than in terms of any assignment of significances to its terms; lacking an interpretation.

UNINTERRUPTIBLE POWER SUPPLY (UPS): A device that includes a battery so that in the event of a power failure it is possible to continue operations for some time in order for a system to be shut down properly so that files are not corrupted or for an alternative power supply to be connected. It also provides protection from power

surges. The **Online UPS** has an inverter that inverts the battery DC supply to AC, so that the equipment is not plugged directly into the domestic power outlet but into the UPS AC outlet.

UNINUCLEATE: Having only a single nucleus, for example, a *uninucleate* yeast cell.

UNION: 1. Also called **sum**, it is the set of all elements that belong to at least one of two or more given sets. It is denoted by the cup symbol $\cup$, the union between two sets S and T is written as $S \cup T$. 2. The growing together of tissues separated by injury, as of the ends of a fractured bone or of the edges of a wound.

UNIPAROUS: 1. Forming only a single axis at each branching from each flowering stem, as occurs in some cymes. 2. Producing only one egg at a time. 3. Producing only one offspring at each birth. 4. Of a woman, having borne only one child.

UNIPETALOUS: Having only a single petal.

UNIPOLAR NEURON: A neuron that has one main process, the axon extending from its cell body resulting from the fusion of two polar processes during development, one branch of the process serving as a sensory nerve fibre and a second branch entering into synaptic contact with neurons in the spinal cord or brainstem.

UNIPORTER: A transport protein that transfers molecules across a membrane in one direction only, down their concentration.

UNIPROT (UNIVERSAL PROTEIN RESOURCE): An international consortium that maintains several databases, serving as a central repository for information about protein sequences and functions. It was established in 2002 to pool the database resources of three organisations to form the basis for the UniProt Knowledge base and associated databases.

UNIQUE: 1. Of or denoting a value, such that it the only one satisfying some condition. For example, the unique positive square root of 4 is 2, although -2 is another square root of 4. 2. **Unique number** is the constant U_n that results if a number A_n consisting of n consecutive digits, in ascending order, is subtracted from the number A_n' obtained by reversing the digits of A_n . For example, a three-digit number 345, if subtracted from its reverse 543, yields a difference of 198. Any other three-digit number subtracted from its reverse gives the same difference. Thus $U_3 = 198$. Similarly for a number with four consecutive digits, the unique number $U_4 = 3087$. The first ten unique numbers are: $U_1 = 0$, $U_2 = 9$, $U_3 = 198$, $U_4 = 3087$, $U_5 = 41976$, $U_6 = 530865$, $U_7 = 6419754$, $U_8 = 75308643$, $U_9 = 864197532$ and $U_{10} = 9753086421$. 3. **Unique up to** for a relation defines elements within some structure that are related by a relative identity relation; identical relative to that relation. For example, positive integers have prime factorisations that are unique up to order means that the index of each prime is unique, but that their order is irrelevant.

UNIQUE FACTORISATION DOMAIN (GAUSSIAN DOMAIN): An integral domain in which every non-zero non-unit is uniquely representable as a finite product of irreducible elements, up to permutation. Since the Gaussian integers are a Gaussian domain and $5 = (1 + 2i)(1 - 2i)$, 5 is reducible in the Gaussian integers; 3 is irreducible therein and hence prime, because it is in a Euclidean domain.

UNIQUE FACTORISATION THEOREM: 1. Also known as **fundamental theorem of arithmetic**, it is the theorem stating that any positive integer can be expressed as a unique product of prime numbers. 2. Any theorem asserting unique factorisation, for example, as in Euclidean domain.

UNIQUE IDENTIFIER (UID): In computer science, it is a numeric or alphanumeric string, such as a product code, registration code and national identification number that is exclusive to that particular entity within a given system.

UNIQUE NUMBER: The constant U_n that results if a number A_n consisting of n consecutive digits, in ascending order, is subtracted

from the number A_n' obtained by reversing the digits of A_n . For example, a three-digit number 345, if subtracted from its reverse 543, yields a difference of 198. Any other three-digit number subtracted from its reverse gives the same difference. Thus $U_3 = 198$. Similarly for a number with four consecutive digits, the unique number $U_4 = 3087$. The first ten unique numbers are:

$U_1 = 0$, $U_2 = 9$, $U_3 = 198$, $U_4 = 3087$, $U_5 = 41976$, $U_6 = 530865$, $U_7 = 6419754$, $U_8 = 75308643$, $U_9 = 864197532$ and $U_{10} = 9753086421$.

UNIQUE QUANTIFIER: In logic, the strengthened existential quantifier, used assert that a predicate is uniquely instantiated, written as $(\exists! x)Fx$ and defined contextually as

$(\exists! x)Fx = (\exists x)(Fx \ \& \ (\forall y)(Fy \rightarrow x = y))$. It is the first member of the sequence of strong numerical quantifiers.

UNIQUENESS THEOREM (UNICITY): The unicity theorem is the uniqueness of a mathematical object, which usually means that there is only one object fulfilling given properties or that all objects of a given class are equivalent (i.e., they can be represented by the same model). The object of many uniqueness theorems is the solution to a problem or an equation; in such cases, a uniqueness theorem is normally combined with an existence theorem.

UNIRAMIA (MANDIBULATA): A phylum of arthropods that contains the chilopoda, diplopoda, pauropoda and symphyla and hexapoda (insects).

UNIRAMOUS APPENDAGE: A type of appendage that is characteristic of insects and other members of the subphylum uniramia. It consists of an unbranched series of segments.

UNISEPTATE: Having only one septum, as in a silicle or silique.

UNISERIATE: Arranged in a single row or series.

BASE QUANTITY	UNIT	SYMBOL
length	metre	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
temperature	kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd

UNISEXUAL: 1. A term describing animals or plants with either male or female reproductive organs but not both. Most of the more advanced animals are unisexual but plants are often hermaphrodite. 2. Having either stamens or pistils but not both.

UNIT: 1. A standard quantity defined and adopted by convention and/or by law, in relation to which other quantities are measured. A **base unit** is one of a set of seven SI units without combination from which all units of physical quantities can be built from. These seven units are called base units because none of them can be expressed as combinations of the other six. Examples of S.I. base units are as follows: metre with the symbol m for length, kilogram with the symbol k for mass, second with the symbol s for time, ampere with the symbol A for electric current, kelvin with the symbol K for temperature, candela with the symbol cd for luminous intensity, and mole with the symbol mol for amount of substance. Some base units are given special names, for example, the kilogram metre per second squared ($\text{kg} \cdot \text{m} \cdot \text{s}^{-2}$) is called the newton (N) and the kilogram metre squared per second squared ($\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$) is called the joule (J). 2. The lowest positive natural number or the first digit to the left of the decimal point in decimal notation, representing a whole number less than ten or a set containing a single number. 3. An amount of an enzyme, hormone, drug or other agent that produces a specific effect, often

as specified by an internationally agreed standard. 4. A component or an assembly of components that performs a specific function. 5. An individual or discrete part of an element into which something can be divided, especially for analysis. 6. A single person, thing or group, usually regarded as a whole part of something larger.

UNIT CELL: The group of smallest particles consisting of atoms, ions or molecules in a crystal that is repeated in three dimensions in the crystal lattice.

UNIT CIRCLE: The loci of points in the plane which are precisely 1 unit from the origin or a circle with a radius of 1. Frequently, especially in trigonometry, the unit circle is the circle of radius one centred at the origin (0, 0) in the Cartesian coordinate system in the Euclidean plane. The unit circle is often denoted S^1 ; the generalisation to higher dimensions is the unit sphere. For example, if (x, y) is a point on the unit circle in the first quadrant, then x and y are the lengths of the legs of a right triangle whose hypotenuse has length 1. Thus, by the Pythagorean theorem, x and y satisfy the following equation:

$x^2 + y^2 = 1$. Since $x^2 = (-x)^2$ for all x and since the reflection of any point on the unit circle about the x - or y -axis is also on the unit circle, the above equation holds for all points (x, y) on the unit circle, not just those in the first quadrant. Because the radius is 1, sine, cosine and tangent can be directly measured. Where $\theta = 0^\circ$, then $\cos \theta = 1$, $\sin \theta = 0$ and $\tan \theta = 0$. If the angle is 90° , $\cos \theta = 0$, $\sin \theta = 1$ and $\tan 90^\circ$ is undefined.

UNIT CUBE: A cube with edge length 1. A unit square has a side length of 1.

UNIT DISK: Any neighbourhood in a metric space, of which the radius is unity; especially the region in the complex plane defined by $\{x : |x| < 1\}$.

UNIT FRACTION: A fraction whose numerator is 1, for example, in the fraction $\frac{1}{4}$, 1 is the unit fraction.

UNIT INTERVAL: The interval between 0 and 1 on a number line.

UNIT MAGNETIC POLE: Two equal magnetic poles of the same sign have unit value when they repel each other with a force of 1 dyne if placed 1 centimetre apart in a vacuum.

UNIT POINT: A point that, together with a given triangle of reference, determines a system of homogeneous coordinates for a two-dimensional algebraic geometry; the unit point is chosen to be linearly independent of any two vertices of the triangle of reference.

UNIT RATE: The simplified value of a ratio whose denominator equals one, for example, 80 kilometres per hour.

UNIT SET (SINGLETON): A set having a single member; an example is the set $\{0\}$.

UNIT SQUARE: A square of side length 1 unit. Often, the unit square refers specifically to the square in the Cartesian plane with corners at (0, 0), (1, 0), (0, 1) and (1, 1).

UNIT SYSTEM: A set of base units, together with derived units, defined in accordance with given rules, for a given system of quantities.

UNIT VECTOR: A vector that has 1 as its magnitude. If a is any non-zero vector the unit vector in the direction of a is given by $\frac{a}{|a|}$ and is denoted by $\hat{a}$.

UNITARY: 1. Consisting of or relating to a single object or individual or undivided and existing as a unit. 2. Any quantity taking or assigned the value one. 3. Also known as **neutral element**, it is the element of a set of which the product with any other element under a multiplicative operation is that other element, its multiplicative identity. 4. The greatest member of a lattice or partially ordered set such as the universal set and denoted by V .

UNITARY EQUIVALENCE: In mathematics, the property of two operators or matrices of being similar with respect to a unitary matrix. Hence $A = UBU^*$, with U unitary and the operators are also congruent. Any Hermitian matrix is unitarily equivalent to a real diagonal matrix

and any normal matrix is unitarily equivalent to a complex diagonal matrix. If the matrix is real the unitary matrix may be supposed real and so orthogonal.

UNITARY MATRIX: An $n \times n$ matrix U is unitary if and only if $\bar{U} = U'$, where $\bar{U}$ is a complex matrix and U is a real matrix. The condition that $\bar{U}^{(-1)} = U'$ is equivalent to the condition that $\bar{U}U' = I_n$. If all the elements of U are real, then this condition becomes $\bar{U}U' = I_n$ and a unitary matrix is orthogonal. Unitary matrices are the complex analogues of orthogonal matrices.

UNITARY METHOD: A method used in problems of proportion and ratio, which involves calculating the value of one unit or item and multiplying by the number of items required.

UNITARY MODULE: A module over a ring with identity in which the product of the ring identity with each element is that element.

UNITARY SPACE (HERMITIAN VECTOR SPACE): A complex vector space on which is defined an inner product.

UNITARY TRANSFORMATION: A linear transformation on a vector space which preserves inner products and norms of the form $\mathbf{o}' = U\mathbf{o}U^{-1}$, where $\mathbf{o}$ is an operator, U is a unitary matrix and U^{-1} is its inverse, i.e. if the matrix obtained by interchanging rows and columns of U and then taking the complex conjugate of each entry, denoted U^+ , the inverse of U is $U^+ = U^{-1}$. The inverse of a unitary transformation is itself a unitary transformation. In quantum mechanics, a unitary transformation corresponds to rotation of axes in the Hilbert space formulation of states. Such a transformation does not alter the state vector, but a given state vector has different components when the axes are rotated. Unitary transformations are important in quantum mechanics.

UNITED NATIONS FRAMEWORK CONVENTION ON CLIMATE CHANGE (UNFCCC): A Convention adopted on 9 May 1992 in New York and signed at the 1992 Earth Summit in Rio de Janeiro by about 195 countries. Its ultimate objective is the "stabilization of greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system."

UNIVALENT (MONOVALENT): 1. Referring to a chemical element or isotope that has a valency of one or capable of replacing a single hydrogen atom in combination. Alkyl groups and hydroxyl ions are **univalent**. Examples are: hydrogen chloride (HCl), chlorine (Cl_2), perchlorate (HClO_4) and sodium hydride (NaH). 2. Referring to a chromosome that remains unpaired during the cell division meiosis that precedes sex cell formation.

UNIVALENT FUNCTION (SCHLICHT FUNCTION; SIMPLE FUNCTION): In mathematics, a complex function on some domain, that is analytic and taking no value more than once in the domain. A schlicht function mapping the finite complex plane onto itself is linear.

UNIVERSAL: In logic, of a statement or proposition, affirming or denying something of every member of a class of objects; containing a universal quantifier. For example, "all men are wicked" or "no pigs can fly" are universal statements.

UNIVERSAL ALGEBRA: The study of relational structures on sets, including groups, rings, fields, lattices, etc.

UNIVERSAL APPROXIMATION: Having the ability to approximate any function to an arbitrary degree of accuracy. Neural nets are universal approximators.

UNIVERSAL ASYNCHRONOUS RECEIVER-TRANSMITTER (UART): A type of integrated circuit used for serial communications over a computer or peripheral device serial port that converts the bytes into serial bits and transmits those bits through an asynchronous transmission, stripping out the start and stop bits for each character.

UNIVERSAL CONSTANTS (FUNDAMENTAL CONSTANTS): Those groups of physical constants which play a fundamental role in the basic theories of physics and whose parameters do not change throughout the universe. Examples are the charge on an electron, the speed of

light in free space, the Planck constant, the gravitational constant, the electric constant and the magnetic constant.

UNIVERSAL COMPUTATION: Capable of computing anything that can in principle be computed; being equivalent in computing power to a Turing machine or the lambda calculus.

UNIVERSAL COMPUTER: A computer that is capable of universal computation, which means that given a description of any other computer or program and some data, it can perfectly emulate this second computer or program. Home PCs are not universal computers because they have only a finite amount of memory. However, in practice, this is usually ignored.

UNIVERSAL GRAVITATIONAL CONSTANT (GRAVITATIONAL CONSTANT; NEWTON'S CONSTANT): The numerical factor relating force, mass and distance in Newton's theory of gravitation that yields the force one body exerts on another when multiplied by the product of the masses of the two bodies and divided by the square of the distance between them. Its symbol is G . The attractive force, F , between two bodies of mass m_1 and m_2 and the distance between them, r , is expressed

as $F = \frac{Gm_1m_2}{r^2}$. The numerical value of G is expressed as

$$G = \frac{Fr^2}{m_1m_2} = 6.67259(85) \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}.$$

UNIVERSAL INDICATOR: A mixture of acid-base indicators that change colour, for example, red-yellow-orange-green-blue over a range of pH values. It is used to determine the acidity or alkalinity of a solution.

UNIVERSAL INSTANTIATION (UNIVERSAL ELIMINATION): In logic, the rule of predicate calculus in accordance with which any instance may be validly inferred from a universally quantified statement; the elimination rule for the universal quantifier.

UNIVERSAL LAW OF GRAVITY: Formula that measures the force of gravity; developed by Sir Isaac Newton. It states that any two objects attract each other with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

UNIVERSAL MOBILE TELECOMMUNICATIONS SYSTEM (UMTS): A third generation (3G) of mobile phone networks based on the GSM standard that transmits broadband data at rates of up to 2 megabits per second (Mbps) or more.

UNIVERSAL MOTOR (AC/DC MOTOR): A series motor that may be operated at approximately the same speed and output on either direct current or single-phase alternating current. It is normally designed for capacities less than 0.75 kW. It is usually operated at high speed of 3500 revolutions per minutes (rpm) loaded and 8000 to 10,000 revolutions per minute unloaded. For lower speeds, reduction gears are often employed, as in the case of electric hand drills or food mixers. As in all series motors, the rotor speed increases as the load decreases and the no-load speed is limited only by friction and windage.

UNIVERSAL PLUG AND PLAY (UPnP): A standard that uses Internet and Web protocol that simplifies computer networking by allowing both wired and wireless devices to communicate without proper configuration and minimal setup by just plugging in. When a user plugs a device into the network, the device configures itself, acquires a TCP/IP address and use a discovery protocol based on the Internet's Hypertext Transfer Protocol (HTTP) to announce its presence on the network to other devices.

UNIVERSAL PROTEIN RESOURCE (UNIPROT): An international consortium that maintains several databases, serving as a central repository for information about protein sequences and functions. It was established in 2002 to pool the database resources of three organisations to form the basis for the UniProt Knowledge base and associated databases.

UNIVERSAL QUANTIFIER: In logic, an operator containing a variable, written (x) or $(\forall x)$ that indicates that the open sentence that follows is true for all members of the domain; for example $(x)(Fx \rightarrow Gx)$ which is read as "for all x , if x is an F then it is a G ", that

is, all F s are G s and is true if and only if $Fa \rightarrow Ga$ is true of every member of the domain.

UNIVERSAL SERIAL BUS (USB): A standard that was introduced in 1995 by Intel, Compaq, Microsoft and several other computer companies that defines the cables, connectors and communications protocols used in a bus for connection, communication and power supply between computers and electronic devices. USB 1.x is an external bus standard that supports data transfer rates of 12 Mbps and is capable of supporting up to 127 peripheral devices. Other types include USB 2.0, also known as hi-speed USB and capable of supporting a transfer rate of up to 480 Mbps and is backwards compatible (it is capable of supporting USB 1.0 and 1.1 devices and cables); and USB 3.0 which is the latest version (2012) of the USB protocol. USB 3.0 has improved performance and speed, including USB thumb drives, digital cameras, external hard drives, MP3 players and other devices. A USB Port is a type of socket into which a plug at one end of the cable can be inserted. The plug at the other end varies according to the device such as a Nokia or Samsung cell phone. USB ports and cables are used to connect hardware such as printers, scanners, keyboards, mice, flash drives, external hard drives, cameras and to computers of all kinds, including desktops, tablets, laptops, netbooks, etc.

UNIVERSAL SERIAL BUS DRIVE (USB DRIVE; USB FLASH DRIVE; DATA STICK; PEN DRIVE; KEYCHAIN DRIVE; THUMB DRIVE; JUMP DRIVE): Any portable drive storage device that can be attached to a computer through a special type of connection. USB drives range in capacity from 256MB to 16GB and offer much more storage than writable CDs and DVDs. They are often used in backup, data transfer, storage of MP3 files, etc.

UNIVERSAL SET: A set that contains all of the possible elements and all of the subsets relevant to the solution of a specific problem.

UNIVERSAL SOLVENT: A material or solvent which instantly dissolves or otherwise changes most other materials. Water is sometimes called the universal solvent, because it can dissolve a large variety of substances, due to its chemical polarity.

UNIVERSAL TIME, UT (UNIVERSAL TIME COORDINATED; UTC): Time based on international atomic time, a uniform time derived from the frequencies of certain atomic transitions and measured by an atomic clock. It is used as the international standard for calculating the time in other zones. It is the name adopted in 1972 for Greenwich Mean Time (GMT), the time scheme set to have 12:00 noon when the Sun is nominally overhead on the Greenwich meridian, but precisely synchronised, at the level of the second, with the atomic-based TAI (Temps Atomique International). The elapsed time from solar noon to noon is not consistent throughout the year; being overhead relates to an average position.

UNIVERSAL VEIL: A membrane that encloses the developing fruiting body of many basidiomycete fungi. It ruptures as the stalk elongates, the remnants forming the volva at the base of the stalk and the flecks of tissue seen on the upper surface of the cap.

UNIVERSALITY: 1. Also known as **background independence**, it is the concept or assumption, fundamental to all physical sciences that the nature of reality is consistent throughout all of space and time. More specifically, no observer can, under any circumstances, perform a measurement that yields a result logically inconsistent with a previous measurement, under a set of rules that are independent of where and when the observations are made. 2. The observation that there are properties for a large class of systems that is independent of the dynamical details of the system. Systems display universality in a scaling limit, when a large number of interacting parts come together. 3. The property of being able to perform different tasks with the same underlying construction just by being programmed in a different way. Universal systems are effectively capable of emulating any other system. Digital computers are universal, but proving that idealised computational systems are universal can be extremely difficult and technical.

UNIVERSALLY MEASURABLE SET: In mathematics, a set E that is measurable for each Borel measure on a topological space. Thus, given a measure μ , there exist $G \subset E \subset F$ with $\mu(F \setminus G) = 0$, where G and F are Borel sets, these being taken as the sigma algebra generated by all closed subsets. In a separable metric space, any Souslin set is universally measurable. Every measurable set is universally measurable, but a universally measurable set is generally not measurable.

UNIVERSE: All space and its contents including stars, planets, galaxies, etc. or all the matter, energy and space that exist, whether known or unknown. Astronomers believe that about 14 billion years ago the universe began as an explosive event resulting in a hot, dense, expanding sea of matter and energy. This event is known as the **big bang**.

UNIVERSE OF DISCOURSE (DOMAIN OF DISCOURSE; UNIVERSE OF INTERPRETATION): In logic, the complete set of individuals able to be referred to or quantified over in an interpreted theory.

UNIX-TO-UNIX COPY PROGRAM (UUCP): In computer science, it is a Unix protocol that allows users to share files, read Usenet articles and exchange e-mail.

UNIX-TO-UNIX NETWORK (UUNET): US-based Internet service provider of Internet access, which was the first commercial Internet Service Provider, founded in 1987.

UNKNOWN: In mathematics, it is a variable, whose value is to be found by solving an equation. For example, the unknowns in the equation $3y = 4x + 5$ are y and x .

UNLEADED PETROL: Petrol manufactured without the addition of tetraethyl lead, also known as tetraethyl with the chemical formula $Pb(C_2H_5)_4$. Tetraethyl lead is a substance added to petrol to reduce or stop faulty fuel combustion associated with a knocking sound. Unleaded petrol has a slightly lower octane rating than leaded petrol, but has the advantage of not polluting the atmosphere with lead compounds.

UNNILENNIUM (MEITNERIUM): A radioactive transactinide element with the atomic number 109, symbol Mt. It was first made in 1982 by Peter Armbruster and a team in Darmstadt, Germany, by bombarding bismuth-209 nuclei with iron-58 nuclei. Only a few atoms have ever been detected.

UNNILHEXIUM (SEABORGIUM): An artificially produced radioactive element with atomic number 106 that is produced by bombarding Californium-249 nuclei with oxygen-18 nuclei, chemical symbol Sg. It is named after the scientist Glenn Seaborg.

UNNIOCTIUM (HASSIUM): A synthetic element, with the symbol Hs, atomic number 108, which is a radioactive transactinide element. It is the heaviest member of the Group 8 elements. It has a half-life of approximately 9.7 seconds.

UNNILENNIUM (DUBNIUM): A radioactive transactinide element, with the symbol Db and atomic number 105. It is in Group 5 (VB) of the Periodic Table and it can be made by bombarding californium-249 nuclei with nitrogen-15 nuclei. Each dubnium atom has a very large nucleus or central mass containing positively charged particles called protons and neutral particles called neutrons. The large number of particles in the nucleus makes the atom unstable and causes the atoms to split apart into smaller components soon after creation. The most stable isotope is ^{268}Db with a half-life of 28 hours. It is the longest lived transactinide isotope, being a reflection of the stability of the $Z = 108$ and $N = 162$ closed shells and the effect of odd particles in nuclear decay.

UNNILENNIUM (RUTHERFORDIUM): An artificially produced radioactive metal with the symbol Rf and atomic number 104. It is made by bombarding an actinide with carbon, oxygen or neon atoms. Its most long-lived isotopes have mass numbers of 253, 255, 257 and

259 with half-lives of 1.8, 1.6, 4.7 and 3.4 seconds, respectively. It was named after Ernest Rutherford (1871-1937), New Zealand-born British physicist, who became known as the father of nuclear physics.

UNNILENNIUM (BOHRIUM): A radioactive transactinide element, with the symbol Bh and atomic number 107. It is the heaviest member of Group VIIB first made in 1981 by Peter Armbruster and a team in Darmstadt, Germany, by bombarding bismuth-209 nuclei with chromium-54 nuclei. Its most stable known isotope, ^{270}Bh , has a half-life of 61 seconds. Chemical experiments have confirmed bohrium's predicted position as a heavier homologue to rhenium with the formation of a stable +7 oxidation state. Only a few atoms of bohrium have ever been detected.

UNPLASTICISED PVC (UPVC): A tough hardwearing form of PVC with a density of 1.4 g/cm^3 . It is cheap, rigid with good resistance against chemicals, sunlight and oxidation from water. It is flame-retardant and comes in various colours and can be transparent. However, it is difficult to process and requires modification. It is used for window frames, for pipes, gutters, window frames and a range of exterior building products. Parts can be welded using a variety of methods including solvents, ultrasonic welding, hot gas and friction.

UNREACTIVE: Failing to react with a specified chemical species under specified conditions. The term should not be used in place of stable, since a relatively more stable species may nevertheless be more reactive than some reference species towards a given reaction partner.

UNRUH EFFECT: The phenomenon that an accelerating body would seem to be surrounded by particles at a non-zero temperature, which is proportional to the acceleration. Also, the vacuum state of a non-accelerating observer is different from that of an accelerating observer, due to distortion of the zero-point fluctuation.

UNS- (UNSYM-): A chemical prefix denoting that the substituents of an organic compound are structurally unsymmetrical with respect to the carbon skeleton or with respect to a function group.

UNSATURATED COMPOUND: A chemical compound in which two adjacent atoms are bounded by a double or triple covalent bond. An example is carbon compounds such as olefins, diolefins and unsaturated fatty acids which possess double or triple bonds between two carbon atoms.

UNSATURATED FAT: A triglyceride with double covalent bonds between some carbon atoms. Fats are designated as saturated or unsaturated, depending on whether the chemical bonds between the carbon atoms of the fat molecules contain all the hydrogen atoms they are capable of holding (saturated) or have capacity for additional hydrogen atoms (unsaturated). Unsaturated fats are liquids, while saturated fats generally are solid at room temperature. Examples of unsaturated fats are cottonseed and peanut oils. Unsaturated fats may be converted to saturated fats by adding hydrogen atoms in a process known as **hydrogenation**. This process creates saturated fats called **trans-fatty acids**. Intake of foods containing more unsaturated fats than saturated fats may contribute to reduced blood cholesterol levels.

UNSATURATED FLOW: 1. The movement of water in a porous medium in which the pore spaces are not filled to capacity with water. 2. Two phase flow through pores only partially filled with water and air.

UNSATURATED HYDROCARBONS: Hydrocarbons that contain at least one double or triple carbon-to-carbon bond that is not in an aromatic ring. Examples are ethylene, propadiene and acetylene.

UNSATURATED SOLUTION: A solution that is capable of dissolving more solute than it already contains at the same temperature. It contains less solute than a saturated solution at a particular temperature.

UNSATURATED ZONE (VADOSE ZONE): The part of the subsurface between the land surface and the groundwater table where groundwater is at atmospheric pressure. Its thickness varies greatly, from being absent to hundreds of metres, depending upon factors such as the environment and the type of earth material present. It stores water, plant nutrients, and other substances and is a zone of

transmission of water and other substances. In this zone the water content is below saturation (for the specific soil).

UNSATURATION: A state in which the atomic bonds of an organic compound's chain or ring are not completely satisfied resulting in a double bond such as for olefins or a triple bond such as for acetylenes.

UNSHIELDED TWISTED PAIR CABLE (UTP): A cable used in computer networking that consists of two shielded wires twisted around each other.

UNSOLVABLE PROBLEM: In computer science, it is a class of problems for which it is impossible to find a single algorithm that always leads to a correct yes or no answer. There are two major types of such problems: problems with no known algorithmic solution and problems whose best known algorithmic solutions require unreasonable amount of time to solve.

UNSTABLE: 1. Tending to decompose or change into other compounds; the chemical species concerned has a higher molar Gibbs energy than some assumed standard. Very unstable chemical species tend to undergo exothermic unimolecular decompositions. Variations in the structure of the related chemical species of this kind generally affect the energy of the transition states for these decompositions less than they affect the stability of the decomposing chemical species. Low stability may therefore parallel a relatively high rate of unimolecular decomposition. Thermodynamic stability occurs when a system is in its lowest energy state or chemical equilibrium with its environment. 2. In physics, it defines an elementary particle or subatomic particles having very short lifetime; or spontaneously decomposing by nuclear decay. 3. An electrical circuit, mechanical body, etc., having a tendency to self-oscillation.

UNSTABLE EQUILIBRIUM: The state of a system that will move further from its original condition after experiencing a slight disturbance.

UNSTABLE ION: An ion which is sufficiently excited to dissociate within the ion source.

UNWINDING PROTEINS: Proteins that help to unwind double-stranded DNA during DNA replication that have a higher affinity for single-stranded DNA than for double-stranded DNA and hold the strands apart during replication.

UPDRAUGHT (UPDRAFT): Up-ward moving of air currents that occur during a thunderstorm. It also occurs as part of the turbulence that is created when air passes over topographic barriers such as mountains.

UPFLOW: An operation in which a solution enters at the bottom of the unit and leave at the top in an ion-exchange unit.

UPHILL TRANSPORT: A process in which diffusion of a component occurs from a less concentrated stream to a more concentrated permeable stream.

UPLAND SOIL: Soil located on elevated land and outside of a flood plain that is not subject to flooding, may be subject to erosion, and may have been subjected to long-term erosion and subsequent loss of a portion of topsoil.

UPLINK: Sending data or information from an Earth station to an airborne station. 1. The subsequences of the elements beyond the n^{th} of a given real sequence, expressed as $\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} [\sup \{a_m : m \geq n\}]$. This produces the largest cluster point of the sequence and may be positive infinity. The sequence has a limit if and only if the limit superior and the limit inferior are identical and in that case the limit is their common value. 2. The set of points that are in infinitely many members of a given sequence $\{A_n\}$, given as $\lim_{n \rightarrow \infty} A_n = \bigcap_{m=1}^{\infty} \left\{ \bigcup_{n \geq m} A_n \right\}$. The sets are said to have a limit if the limit superior and limit inferior, agree and the limit is then this common set.

UPLOAD (U/L): A term used to describe the process of transferring a file to another computer through a modem or network. Commonly

uploaded programs include text file, an image file, a sound file or a video file. Upload is the opposite of **download**, which is the process of retrieving data from another computer, either over a network or modem.

UPPER: 1. Indicating a limit or bound of a set of numbers greater than or equal to every member of the set. 2. Referring to later in a particular geologic period, formation or system.

UPPER ATMOSPHERE: The part of the Earth's atmosphere above the troposphere. It is beyond the reach of balloons. The atmosphere below the stratosphere is called the lower atmosphere.

UPPER BOUND (BOUNDARY): Maximum level of any mathematical function. An upper bound of a subset S of some partially ordered set $(P, \leq)$ is an element of P which is greater than or equal to every element of S .

UPPER CRITICAL TEMPERATURE (CRITICAL THERMAL MAXIMUM): The maximum body temperature that can be tolerated by given species above which cellular compounds and processes become disrupted and the organism eventually dies. It ranges between 30 °C (303 K) to 45 °C (318 K) for most animals; few plants can survive leaf temperature of 50 °C (323 K). Some desert plants can withstand a temperature of 60 °C (333 K) and above.

UPPER FLAMMABLE LIMIT (UPPER EXPLOSIVE LIMIT): The maximum percentage of flammable gas or vapour in the air capable of producing a flash of fire in presence of an ignition source. Above this, ignition cannot take place because the ratio of the gas to oxygen is too high.

UPPER INTEGRAL (UPPER DARBOUX INTEGRAL): The infimum of the sum of products, as the lengths of the subintervals $[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n]$ (where $a = x_0 < x_1 < \dots < x_n = b$) tend to zero of a function on the interval $[a, b]$. It is greater than or equal to the lower Darboux integral (a common value for the lower and upper Darboux integrals). If it exists that the upper Darboux integral is equal to the lower one then the function is Riemann integrable.

UPPER LIMIT (LIM SUP; LIMIT SUPERIOR): 1. the limit as n tends to infinity of the suprema of the subsequences of the elements beyond the n^{th} of a given real sequence, expressed as $\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} [\sup \{a_m : m \geq n\}]$. This produces the largest cluster point of the sequence and may be positive infinity. The sequence has a limit if and only if the limit superior and the limit inferior are identical and in that case the limit is their common value. 2. The set of points that are in infinitely many members of a given sequence $\{A_n\}$, given as $\lim_{n \rightarrow \infty} A_n = \bigcap_{m=1}^{\infty} \left\{ \bigcup_{n \geq m} A_n \right\}$. The sets are said to have a limit if the limit

superior and limit inferior, agree and the limit is then this common set.

UPPER LIMIT OF MEASUREMENT: Highest value of the air quality characteristic, which can be measured by an instrument. Its variations, caused, for example, by instability are expected to lie within specified limits. The difference between the lower detection limit and the upper limit of measurement constitutes the dynamic range of the instrument.

UPPER QUARTILE: In mathematics, in an ordered data set, the middle value of the data above the median. It is the median of the upper half of a data set. Data values at the median are not included when finding the upper quartile.

UPPER SUM (UPPER DARBOUX SUM): The weighted sum of the products of the supremal values of a function on a succession of subintervals of a given interval with the lengths of the subintervals.

UPPER TRIANGULAR MATRIX: A square matrix having only zero entries below the main diagonal. An example is

$$A = \begin{bmatrix} 2 & 7 & 1 \\ 0 & 4 & 6 \\ 0 & 0 & 5 \end{bmatrix}$$

UPPER-TAIL PROBABILITY: Probability that the random variable X takes values greater or equal to x , where x is greater than the expectation of X .

UPS (UNINTERRUPTIBLE POWER SUPPLY): A device that includes a battery so that in the event of a power failure it is possible to continue operations for some time in order for a system to be shut down properly so that files are not corrupted or for an alternative power supply to be connected. It also provides protection from power surges. The **Online UPS** has an inverter that inverts the battery DC supply to AC, so that the equipment is not plugged directly into the domestic power outlet but into the UPS AC outlet.

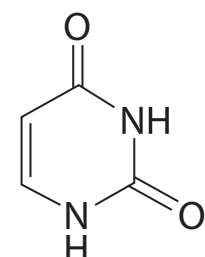
UPTAKE: 1. In biology, it is the transfer of substances from the environment to plants, animals, and humans. It is the process by which a substance crosses an absorption barrier and is absorbed into the body. 2. In climate change, it is the addition of a substance of concern to a reservoir. The uptake of carbon-containing substances, in particular carbon dioxide, is often called **carbon sequestration**.

UPTHRUST: An upward force on an object fully or partially submerged in a fluid or the force that pushes an object up and makes it seem to lose weight in a fluid. The upthrust or buoyancy, keeps swimmers on top of the water and keeps ships afloat. A fluid will exert a force upward on a body if it is partly or wholly submerged within it. This is because the deeper into a fluid it goes, the greater the weight of it and so the greater the pressure. According to Archimedes' Principle, the upthrust on an object in a fluid is equal to the weight of the fluid displaced. It is the volume of the object multiplied by the density of the fluid.

UPWELLING: The oceanographic process by which dense, cooler and often nutrient-rich water is brought up from a lower depth to the surface layers of oceans and seas. Many of the world's best fishing grounds are located at such points. Coastal upwelling is often the result of persistent surface winds blowing parallel to the coastline.

URACIL (PYRIMIDINE-2,4(1H,3H)-DIONE; 2,4-DIHYDROXYPYRIMIDINE; 2,4-PYRIMIDINEDIOL): A pyrimidine derivative which is one of the major component bases of nucleotides and the nucleic acid RNA. It consists of a ring containing both nitrogen and carbon atoms.

Its chemical formula is $C_4H_4N_2O_2$ with a density of 1.32 g/cm^3 and a melting point of 335°C (608 K). It occurs in combined form in many important biological molecules, including RNA and several coenzymes active in carbohydrate metabolism. During synthesis of an RNA strand from DNA, uracil pairs with adenine.



Chemical Structure of Uracil

URAEMLA (UREMIA; AZOTAEMIA; AZOTEMIA): A toxic condition of blood poisoning caused by the accumulation in the blood of products that are normally eliminated in the urine. Some of the symptoms are confusion, fits, loss of appetite, nausea, haemorrhagic episodes and coma.

URANINITE: A dark-coloured, dense and radioactive mineral with the chemical formula UO_2 and Mohs scale hardness of 5-6. It is the chief ore of uranium and contains small amounts of radium, thorium, polonium, lead and helium. It is known as **pitch blende** when it occurs in a massive form with a pitchy lustre.

URANIUM: A heavy, lustrous, silvery-white metallic element, malleable, ductile, radioactive and toxic metallic element with the symbol U , atomic number 92, atomic weight 238.03, melting point $1,132^\circ\text{C}$ (1405 K), boiling point $3,818^\circ\text{C}$ (4091 K) and relative density 18.95. It has 14 known isotopes of which $U-238$ is the most abundant in nature. It combines readily with most elements to form compounds that are extremely poisonous. The element occurs in several minerals, including uraninite and carnotite, from which it is extracted and

processed for use in research, nuclear fuels and nuclear weapons.

URANIUM ENRICHMENT: The process by which the percentage composition of uranium-235 is increased through the process of isotope separation in order to make nuclear fission occur.

URANIUM SERIES (URANIUM DECAY SERIES; URANIUM-RADIUM SERIES): A group of isotopes including isotopes of uranium, thorium, protactinium, radium, radon, plutonium, lead, bismuth and thallium with mass number $4n+2$, where n is an integer, representing various stages of radioactive decay in which the heavier members of the group are transformed into successively lighter ones, the lightest being stable.

URANIUM-LEAD DATING: A radiometric dating method used to determine the age of an uranium-containing mineral by measuring the level of lead isotopes produced by the decay of the radioisotopes Uranium -238 (^{238}U) to Lead -206 (^{206}Pb) or the decay of Uranium -235 (^{235}U) to Lead -207 (^{207}Pb). One method of uranium-lead dating depends on measuring the ratio of the amount of helium trapped in the rock to the amount of uranium present while the other method involves measuring the ratio of radiogenic lead present to nonradiogenic leads, with both methods giving reliable results for the ages in the 10^7 - 10^9 years.

URANIUM(IV) OXIDE (URANIUM DIOXIDE): A highly toxic, radioactive, spontaneously inflammable black crystalline solid with the chemical formula UO_2 , occurring naturally as uraninite. Its melting point is $3,000^\circ\text{C}$ (3273 K) with a density of 10.97 g/cm^3 . Uranium dioxide is produced by reducing uranium trioxide with hydrogen: $UO_3 + H_2 \rightarrow UO_2 + H_2O$ at 700°C (973 K). It is used in photographic chemicals, ceramics, pigments and packing of nuclear fuel rods. This reaction takes part in the reprocessing of nuclear fuel and enrichment of uranium for nuclear fuel.

URANIUM(VI) FLUORIDE (URANIUM HEXAFLUORIDE): A highly toxic, radioactive, corrosive, colourless crystal with the chemical formula UF_6 , density 5.09 g/cm^3 (solid), melting point 64°C (337 K), boiling point 56.5°C (329.65 K) and sublimes. It is soluble in carbon tetrachloride, fluorocarbons and liquid halogens. It reacts vigorously with alcohol, water, ether and most metals and it sublimes. It is used in the separation of uranium isotopes by gas diffusion.

URANIUM(VI) OXIDE (URANYL OXIDE; URANIUM TRIOXIDE; URANIC OXIDE): A highly toxic, radioactive orange powder with the chemical symbol UO_3 , density 5.5 - 8.7 g/cm^3 , melting point 200 - 650°C (573 - 923 K) and decomposes. The solid may be obtained by any of the following ways: (i) heating uranyl nitrate to 400°C (673 K); (ii) oxidising triuranium octoxide (U_3O_8) at 500°C (773 K) with oxygen; (iii) sodium diuranate is converted to Uranium(VI) oxide in the enrichment of uranium. Uranium(VI) oxide is used in uranium refining, as a pigment and in ceramics.

URANUS: The name of the seventh planet from the Sun and also the seventh smallest planet in the solar system, revolving about the Sun every 84.01 years at a mean distance of approximately 2.9 billion kilometres. It has a mean equatorial diameter of 51,118 kilometres and a mass 14.6 times that of Earth.

URANYL ACETATE: A widely used electron stain suitable for high-resolution transmission electron microscopy. It stains nucleic acids and proteins thus enhancing the contrast obtained in observations of membranes and membrane-bound organelles.

URBAN HEAT ISLAND (UHI): An urban area that is significantly warmer than its surrounding rural areas due to human activities. The main cause of the urban heat island effect is from the modification of land surfaces. For example, in urban areas, vegetation is cleared for development, but in rural areas most of the region is covered with plants which take up water from the ground through their roots and store the water in their stems and leaves, turning the liquid water into water vapour which is released into the air, thereby cooling

the surrounding areas; a process called transpiration. A secondary contributor to urban heat island is waste heat generated by energy usage.

URCEOLATE: Resembling a pitcher; hollow and contracted near the mouth like a pitcher or urn. **Urceolus** is any structure resembling a small pitcher.

UREA (CARBAMIDE; CARBONYLDIAMIDE; CARBONYLDIAMINE; DIAMINOMETHANAL; DIAMINOMETHANONE): An organic compound with the chemical formula $\text{CH}_4\text{N}_2\text{O}$,

density 1.32 g/cm^3 and a melting point of $133\text{--}135^\circ\text{C}$ ($406\text{--}408 \text{ K}$). It is solid, colourless and odourless, although the ammonia which it gives off in the presence of water, including water vapour in the air, has a strong odour. It is highly soluble in water and non-toxic. It is disposed off mainly in the urine and also in traces of sweat in mammals. It is formed in the liver of mammals by decomposition of unwanted proteins or nitrogen compounds. It is also produced industrially from ammonia and carbon dioxide. It is used in urea-formaldehyde resins and pharmaceuticals, as nitrogen fertiliser and as a source of nonprotein nitrogen for ruminant livestock.

UREA CYCLE (KREBS-HENSELEIT CYCLE; KREBS ORNITHINE CYCLE; KREBS UREA CYCLE): A metabolic pathway in the liver that leads to the synthesis of urea from amino groups and CO_2 . The function of the pathway is to convert the ammonia resulting from catabolism to a non toxic form, which is subsequently excreted in solution in urine or in traces of sweat. The key reaction is the hydrolysis of arginine by arginase to ornithine and urea.

UREA-FORMALDEHYDE RESIN (UREA-METHANAL): A colourless, non-inflammable, weather-resistant thermosetting plastic, made by copolymerising urea with formaldehyde (methanal) or its polymers. It is used in adhesives for bonding plywood and other wood products, moulded objects and medium density fireboards.

UREASE: An enzyme that occurs in plants such as soya beans and acts as a catalyst for the hydrolysis of urea to ammonium and carbon dioxide.

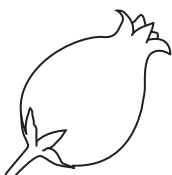
UREDOSPORE (UREDIOSPORE): A dikaryotic spore formed in a spore cluster (a uredosorus or uredium) by certain rust fungi. The uredospores of *Puccinia graminis* are formed on the main host, wheat and are responsible for the characteristic rusty streaks of infected plants. On release they infect more wheat plants.

URLEMENT: A mathematical object which is not a set, but may be an element of a set.

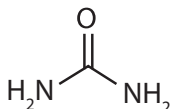
UREOTELIC: Animals that excrete nitrogenous waste in the form of urea. These animals have high liver levels of arginase. Most terrestrial vertebrates such as humans, dogs, rat, sheep, cattle and pigs are ureotic, converting ammonium ions formed during the breakdown of amino acid into urea.

URETHANE: 1. A highly toxic colourless or white crystalline inflammable organic compound with the chemical formula $\text{CO}(\text{NH}_2)\text{OC}_2\text{H}_5$. It is used in veterinary medicine, biochemical research and as a solvent and chemical intermediate. 2. Any of several esters, other than the ethyl ester of carbamic acid.

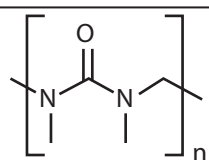
URETHANE RESINS (POLYURETHANES): Synthetic resins containing the repeating group $-\text{NH}-\text{CO}-\text{O}-$. There are numerous types made by copolymerising isocyanate esters with polyhydric alcohols. They have a variety of uses in plastics, paints and solid foams.



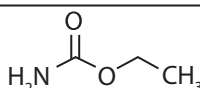
Urceolate



Chemical Structure of Urea

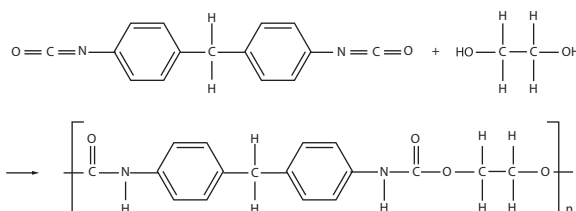


Chemical Structure of Urea-formaldehyde Resin



Chemical Structure of Urethane

URETER: Any of the two tubes made of smooth muscle fibres that carry urine from any kidney to the bladder or cloaca. In adults, the ureters are usually $25\text{--}30 \text{ cm}$ long.



Chemical Structure of Urethane Resins

URETHRA: The duct that carries urine from the bladder to the outside of the body of males and females. In males, it also carries semen.

URI (UNIFORM RESOURCE IDENTIFIER): A string of characters that identifies the name and location of a file or resource in a uniform format. URIs provide a standard way for resources to be accessed by other computers across a network or over the World Wide Web. They are used by software programs such as Web browsers and P2P file-sharing programs to locate and download files. The most common form of URI is the uniform resource locator (URL)

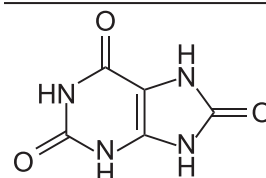
URIC ACID: A white crystalline slightly soluble organic compound with the chemical formula $\text{C}_5\text{H}_4\text{N}_4\text{O}_3$ and melting point of 300°C (573 K).

It is the end product of purine breakdown and is the chief nitrogenous component of the urine in birds, terrestrial reptiles and insects. Uric acid is only slightly soluble in water and is insoluble in alcohol and ether. When heated, the acid forms urea, ammonia and carbon dioxide. It is present in small amounts in human urine and in large amounts in the urine of birds and reptiles. Kidney stones formed by salts of uric acid occur in persons with high levels of uric acid in the urine; and crystals of uric acid accumulate in the synovial fluids of the joints of people, causing gout.

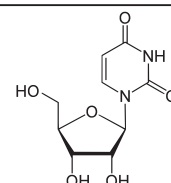
URICOTELIC: Animals that excrete uric acid as the chief component of nitrogenous waste. Uricotelic animals include birds and reptiles.

URIDINE: A solid nucleoside consisting of one uracil molecule attached to a D-ribose sugar molecule. Its molecular formula is $\text{C}_9\text{H}_{12}\text{N}_2\text{O}_6$, molar mass $244.20 \text{ g mol}^{-1}$, density $.99308 \text{ g/cm}^3$ and melting point 167.2°C . The derived nucleotide uridine diphosphate (UDP) is very useful in carbohydrate metabolism. Some foods that contain uridine in the form of RNA include sugarcane extract, tomatoes (0.5 to 1.0 g uridine per kilogram dry weight), brewer's yeast (1.7% uridine by dry weight), beer, broccoli, liver, pancreas, etc. However, virtually none of the uridine in this form is bioavailable. As shown by Handschumacher's Laboratory at Yale Medical School in 1981, it is destroyed in the liver and gastrointestinal tract and no food, when consumed, has ever been reliably shown to elevate blood uridine levels. In infants, consuming mothers' milk or commercial infant formulas, uridine is present as its monophosphate, UMP and this source of uridine is indeed bioavailable.

URINARY SYSTEM: The bodily system of organs and tissues that removes nitrogenous waste products and excess water from the bodies of animals through osmoregulation and excretion. In vertebrates it consists of a pair of kidneys, which produce urine; ureter which drains the kidneys; a bladder which stores the urine; and an urethra that carries the urine to the outside of the body of the mammal.



Chemical Structure of Uric Acid



Chemical Structure of Uridine

URINE: The aqueous fluid of organic and inorganic substances, mostly waste products of metabolism formed by the nephrons of the kidney of animals for the removal of metabolic waste. It is a solution of urea and mineral salts, proteins and waste products in the form of urea, pigment and some acid. It is stored in the bladder before being discharged through the urethra or cloaca. Humans produce about 1.0-1.5 litres per day.

URINOGENITAL SYSTEM (GENITOURINARY SYSTEM): The organ system of the reproductive organs and the urinary system. In mammals, the urinary system structures include kidneys, ureters, bladder, and urethra. The major structures of the reproductive system in males are the testes, sperm ducts, urethra, and penis; in females, they are the ovaries, fallopian tubes, uterus, and vagina.

URL (UNIFORM RESOURCE LOCATOR; WEB ADDRESS): A standardised naming convention for addressing a specific Web site or file on the Internet. It usually consists of the access protocol (http), the domain name (www.hmco.com) and optionally the path to a file or resource residing on that server (trade). It does not have spaces or certain other characters and uses forward slashes to denote different directories. Examples of URLs are <http://www.bro.com/>, <http://web.ust.edu/> and <ftp://info.apple.com/>. Not all URLs begin with "http" and some portions of the URL are not required and many times can be left out.

UROCHORDATA (TUNICATA): A marine subphylum of the phylum Chordata invertebrate chordates in which the body is typically enclosed in a protective tunic. They have the following characteristics: (i) a perforated pharynx or branchial sac, which is used for food collection; (ii) a dorsal notochord restricted to the tail of the larva and the adult in one class; (iii) absence of mesodermal segmentation or a recognisable coelom; (iv) secretion of an outer covering (the test or tunic) which contains large amounts of polysaccharides related to cellulose.

URODELA (CAUDATA): An order of tailed amphibians comprising the newts and salamanders.

UROHIDROSIS: A cooling mechanism practised by some birds, in which they release faeces or urine onto the scaly portions of their legs.

UROLOGY: Medical speciality that is concerned with diseases of the urinary tract of females and males and the male reproductive organs. An **urologist** is the medical expert that specialises in diseases of the urinary organs in females and the urinary and the male reproductive organs.

URONIC ACIDS: A class of sugar acids in which the carbon atom carrying the primary hydroxyl group (i.e. the CH₂OH end of the molecule) is oxidised to a carboxyl group (COOH). Uronic acids are biologically the most important group of the sugar acids and are common constituents of polysaccharides. For example, glucuronic acid, derived from glucose, is a common constituent of gums and mucilages. Galacturonic acid is a component of pectic substances, while mannuronic acid is found in the seaweed gum, alginic acid.

UROPYGIAL GLAND (PREEN GLAND): The oil-secreting gland found on the back, at the base of the tail in most birds; used for preening (cleaning and arranging feathers using the beak) which is part of feather care activities.

URTICACEAE: A family of flowering plants in the nettle family (order Rosales) comprising about 54 genera and 2,625 species of herbs, shrubs, small trees, and a few vines, distributed primarily in tropical regions. These include silkplant (*Boehmeria*); Ramie (*Boehmeria nivea*, commonly known as China grass, white ramie, green ramie and rhea); *Boehmeria cylindrica*, with common names false nettle and bog hemp; and Chinese money plant (*Pilea peperomioides*). Some species possess some medicinal properties, for example, *Urtica urens* (with common names annual nettle, dwarf nettle and small nettle) is used to treat ailments such as asthma, bronchitis, sinusitis, diabetes, gout, cancer, hay fever, hypertension, heart attacks, ulcers, intestinal inflammation, and haemorrhoids.

U.S. CUSTOMARY SYSTEM OF MEASUREMENT: The measuring system used most often in the United States which was developed

from British units. Units for length include inch, foot, yard, and mile; units for weight include ounce and pound; units for volume or capacity include cup, pint, quart, gallon, and cubic units; and the main unit for temperature change is degrees Fahrenheit. However, for scientific and technological reasons, The International System of Units (S.I.), is preferred for many uses by the U.S. National Institute of Standards and Technology (NIST).

USB (UNIVERSAL SERIAL BUS): A standard that was introduced in 1995 by Intel, Compaq, Microsoft and several other computer companies that defines the cables, connectors and communications protocols used in a bus for connection, communication and power supply between computers and electronic devices. USB 1.x is an external bus standard that supports data transfer rates of 12 Mbps and is capable of supporting up to 127 peripheral devices. Other types include USB 2.0, also known as hi-speed USB and capable of supporting a transfer rate of up to 480 Mbps and is backwards compatible (it is capable of supporting USB 1.0 and 1.1 devices and cables); and USB 3.0 which is the latest version (2012) of the USB protocol. USB 3.0 has improved performance and speed, including USB thumb drives, digital cameras, external hard drives, MP3 players and other devices. A USB Port is a type of socket into which a plug at one end of the cable can be inserted. The plug at the other end varies according to the device such as a Nokia or Samsung cell phone. USB ports and cables are used to connect hardware such as printers, scanners, keyboards, mice, flash drives, external hard drives, cameras and to computers of all kinds, including desktops, tablets, laptops, netbooks, etc.

USB DRIVE (UNIVERSAL SERIAL BUS DRIVE; USB FLASH DRIVE; DATA STICK; PEN DRIVE; KEYCHAIN DRIVE; THUMB DRIVE; JUMP DRIVE): Any portable drive storage device that can be attached to a computer through a special type of connection. USB drives range in capacity from 256 MB to 16 GB and offer much more storage than writable CDs and DVDs. They are often used in backup, data transfer, storage of MP3 files, etc.

USB HUB (UNIVERSAL SERIAL BUS HUB): A device that has many sockets on an equipment such as computers, keyboards, monitors, or printers that increases the number of USB ports so that there are more ports available to connect devices to a host system.

USB SMART DRIVE (SMART DRIVE): A USB flash drive having software to interact with a computer or other host device.

USB ON-THE-GO (USB OTG; OTG): An extension of the Universal Serial Bus (USB) that enables peripheral devices to be connected directly to each other without the use of a computer. For example, a mobile phone can act as a host, allowing other external peripheral devices such as flash drives, mice or keyboards to be connected to it.

USENET (USERS' NETWORK): The world's largest bulletin board system, which brings together people with common interests to exchange views and information. It consists of e-mail messages and articles organised into newsgroups. Usenet is uncensored and governed by the rules of netiquette.

USER DATAGRAM PROTOCOL (UDP): An alternative protocol to the TCP/IP that runs on top of IP and is sometimes referred to as UDP/IP. Unlike TCP/IP, UDP does not divide each transmission into packets, which allows for a faster transmission. However, UDP does not provide error checking.

USER INTERFACE (UI; INTERFACE): The portion of a software program or hardware device that is used to navigate and control aspects of the software program or hardware device. A good user interface provides a user-friendly experience, allowing the user to interact with the software or hardware in a natural and intuitive way. Nearly all software programs have a graphical user interface (GUI).

This means the program includes graphical controls, which the user can select using a mouse or keyboard. A typical GUI of a software program includes a menu bar, toolbar, windows, buttons and other controls.

USER-FRIENDLY: A computer system that is easy to use, operate and understand, particularly for those with little understanding or familiarity with computers. It is equally good for experienced users, as user-friendly programs are much easier to learn.

USERNAME (ACCOUNT NAME; LOGIN ID; USER ID): A name, almost always paired with a password that uniquely identifies someone on a computer system. Usernames allow multiple users to use the same computer and also commonly allow users to have their own personal settings and files. Usernames used on Internet websites are commonly used to allow a user to have his or her own personal settings and identification on that website.

UTC (UNIVERSAL TIME COORDINATED; UNIVERSAL TIME, UTC): Time based on international atomic time, a uniform time derived from the frequencies of certain atomic transitions and measured by an atomic clock. It is used as the international standard for calculating the time in other zones. It is the name adopted in 1972 for Greenwich Mean Time (GMT), the time scheme set to have 12:00 noon when the Sun is nominally overhead on the Greenwich meridian, but precisely synchronised, at the level of the second, with the atomic-based TAI (Temps Atomique International). The elapsed time from solar noon to noon is not consistent throughout the year; being overhead relates to an average position.

UTERUS (WOMB): The hollow muscular organ in the pelvic cavity of female mammals in which the embryo develops during gestation. It is situated between the bladder and rectum and is connected to the vagina by the cervix. It is single in humans but paired in most mammals. It has a thick spongy layer for fertilised eggs and thick muscular outer walls which by contraction forces mature foetus outside through the vagina. Most animals that lay eggs, such as birds and reptiles, have an oviduct instead of an uterus.

UTILITY (UTILITIES): Computer software programs that add functionality to a computer or help a computer perform better. Examples are antivirus, backup, disc repair, file management, security and networking programs. Utilities can also be applications such as screensavers, font and icon tools and desktop enhancements. Some utility programs help keep a computer free from unwanted software such as viruses or spyware, while others add functionality that allows customising a desktop and user interface.

UTILITY FUNCTION: In mathematics, it is an increasing, often continuous, real-valued function defined on a set and inducing or corresponding to a preference order by having $u(x) \leq u(y)$ exactly when y is preferred to x . Often one requires that u be quasi-concave as this corresponds to the 'law of diminishing returns'.

UTRICULUS (UTRICLE): The larger of two fluid-filled sacs in the labyrinth of the inner ear, into which the semicircular canals open, the smaller being the sacculus. It has patches of sensory epithelium whose function is to send impulses to the brain indicating the position of the head. 2. In botany, it refers to a thin-walled one-seeded, more or less bladderly-inflated fruit.

UUCP (UNIX-TO-UNIX COPY PROGRAM): In computer science, it is a Unix protocol that allows users to share files, read Usenet articles and exchange e-mail.

UUENCODE (UUENCODE/UUDECODE): In communications, it is a utility program for encoding and decoding messages that converts a binary file into ASCII text that can be exchanged between users in e-mail or Usenet messages.

UUNET (UNIX TO UNIX NETWORK): US-based Internet service provider of Internet access, which was the first commercial Internet Service Provider, founded in 1987.

UV (ULTRAVIOLET): Of or relating to electromagnetic radiation that has a wavelength shorter than that of the violet end of the visible spectrum but longer than that of X-rays. It has frequencies between 790 terahertz and 30 petahertz and wavelengths between 380 and 10 nanometers. UV light is invisible to the human eye. The main source of UV is the Sun; some artificial light such as from tanning and electric arc lamps also radiate UV. UV radiation can burn the skin and cause skin cancer. However, it has some health benefits, because when it hits human skin, it triggers the production of vitamin D, which promotes the growth of bones and teeth.

UV STABILISER: A substance added to a sample to prevent photodeterioration by ultraviolet (UV) light.

UVEA: The middle layer of the vertebrate eye. The uvea consists collectively of the iris, the choroid and the ciliary body. The anterior (front) uvea includes the iris and ciliary body. The posterior (back) uvea is the choroid.

UVULA: The soft central portion of the soft palate which hangs at the back of the pharynx and forms part of the occluding mechanism which can functionally separate the nasopharynx from the pharynx.

ADVERT SPACE

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v: 1. In physics, it is the symbol for velocity. 2. Symbol for the number 5 in Roman numerals. The uppercase letter may also be used.

V: In physics, it is the symbol for volt, the SI unit of electromotive force (emf) or potential difference (pd), which is equal to the potential difference between two points in an electrical circuit when a current of 1 ampere flowing between them dissipates a power of 1 watt.

VACCINE: 1. An agent containing antigens produced from killed, attenuated or live pathogenic microorganisms, synthetic peptides or by recombinant organisms, used for stimulating the immune system of the recipient to produce specific antibodies providing active immunity and/or passive immunity against a particular disease. It is introduced into the body, usually orally, by a hypodermic syringe or by nasal spray. The method of introduction into the body is called **vaccination**, **immunisation** or **inoculation**. The first vaccine, introduced by Edward Jenner in 1798 was against smallpox. Since then, vaccines have been developed against bacterial diseases such as typhoid, whooping cough and tuberculosis and viral diseases such as measles, influenza, rabies and poliomyelitis. Vaccines have dramatically reduced the incidence of a number of deadly diseases and eradicated smallpox in 1980 and also eliminated other diseases such as diphtheria, poliomyelitis and neonatal tetanus. 2. A software program designed to detect and stop the progress of computer viruses.

VACUOLE: A fluid-filled cavity in the cytoplasm of certain unicellular organisms containing items such as water, nutrients and sugars in solution. It also stores the blue, red and purple pigments that give certain flowers their colours. Animal cells usually have several small vacuoles while plant cells usually have one large vacuole bounded by a single-layered membrane called a vacuole membrane or tonoplast. Vacuoles are particularly important in maintaining turgor in plant cells.

VACUOLE MEMBRANE (TONOPLAST): The single semi permeable membrane that bounds the vacuole of plant cells and separates the fluid-filled vacuole from the surrounding cytoplasm.

VACUOUS: In logic, it refers to an operator or expression that is idle and having no import. For example, in (x) (John is tall) the universal quantifier (x) is vacuous.

VACUUM: A region which is theoretically completely empty of matter or a space in which there is a low pressure of gas or relatively few atoms or molecules or a space from which all air or gas has been extracted. A perfect vacuum would have no molecules or atoms but this is practically not achievable as all the materials that surround such a space have a finite vapour pressure. In a soft or low vacuum the pressure is reduced to about 10^{-2} pascal, whereas a hard or high vacuum has a pressure of 10^{-2} to 10^{-7} pascal. Pressure, which is reduced to below 10^{-7} pascal is known as an **ultrahigh vacuum**.

VACUUM ACTIVITY: Patterns of animal behaviour that occur in the absence of external stimuli that normally elicit them. For example, squirrels that have live in metal cages without bedding do all the actions that a wild squirrel does when burying a nut, even though there is no nut to bury.

VACUUM DISTILLATION (REDUCED-PRESSURE DISTILLATION): Liquid distillation under reduced pressure. This helps to lower the boiling point and reduces the risk of thermal dissociation or degradation.

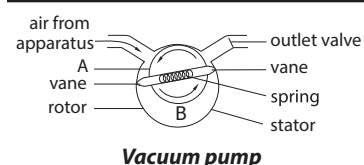
VACUUM FLASK (THERMOS FLASK): A container designed to maintain the desired temperature of the contents. It has two silvered glass walls with a vacuum between its inner and outer glass walls and a metal or plastic outer cage.

VACUUM PAN: A vessel equipped with a vacuum pump used for rapid evaporation by boiling a substance at a low temperature under reduced pressure. It is used to remove water from milk and also used in sugar making for boiling down syrup.

VACUUM PERMITTIVITY (PERMITTIVITY OF FREE SPACE; PERMITTIVITY OF VACUUM): The constant that defines the strength of the electric force between electric charges in vacuum or empty space. In other words, it defines how the electric field reacts to the presence of charges in empty space. Its symbol is ϵ_0 . Mathematically, it is expressed as $\epsilon_0 = 1/(\mu_0 c^2)$, where μ_0 is the magnetic permeability of free space and c is the speed of light in vacuum. Its value is approximately $8.854 \times 10^{-12} \text{ F m}^{-1}$ (farads per meter).

VACUUM PHOTOTUBE: A photo-emissive detector inside an evacuated envelope with a transparent window, the photocathode and the anode. The photocathode can be opaque or semitransparent.

VACUUM PUMP: A pump that reduces the gas pressure in a container or a device that creates partial vacuum, used to pump air, water vapour organic and inorganic solvents and acids. When gas in a closed container is lowered from atmospheric pressure, the operation constitutes an



Vacuum pump

increase in vacuum in this container. Vacuum pumps are divided into two main groups: gas transfer or gas capture. Transfer pumps force gas molecules in a preferred direction by positive displacement or momentum exchange. Ultimately, the gas is compressed until slightly above atmospheric pressure when it is ejected into the atmosphere. They are generally used for high gas loads. Capture pumps immobilise gas molecules on special surfaces within the vacuum system and used to produce oil-free vacuums and UHV pressures. There are many different types of vacuum pumps, which include the liquid ring vacuum pump; rotary piston vacuum pumps; the rotary vane vacuum pump also known as a sliding vane pump; the diffusion pump; the ion pump; the cryogenic pump; etc. The diagram shows the rotary vane vacuum pump (sliding vane pump). In this pump, a slotted vane is placed eccentrically (slightly off-centre) in a cylindrical stator of the pump and rotates with a drum (rotor). As the rotor turns, the sliding vanes are forced out against the casing wall by centrifugal force. The volume of chamber A increases and the pressure decreases and air or gas is forced inside from the apparatus to be evacuated by the inlet pressure. The air in chamber B is then compressed and expelled through a spring-loaded exhaust valve.

VACUUM STATE: The ground state in a relativistic quantum field theory with the lowest possible energy. In quantum mechanics, the vacuum state has a zero-point energy and contains no physical particles, giving rise to vacuum fluctuations, which has significant consequences in quantum electrodynamics.

VACUUM SEALING: A type of packaging process used to slow down deterioration of food sources by reducing atmospheric oxygen, creating an anaerobic environment that limits the growth of aerobic bacteria or fungi, and prevents the evaporation of volatile components.

VACUUM SYSTEM: 1. In urology, it is a mechanical system used to facilitate and maintain an erection. 2. In chemistry it consists of components used to lower the pressure within a mass spectrometer and includes not only the various pumping components but also valves, gauges and associated electronic or other control devices, the chamber in which ions are formed and detected and the vacuum envelope.